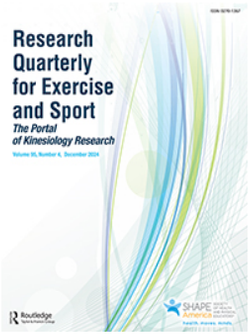

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National Football League Scouting Combine Test Results of Elite-Level National Football League Players

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ABSTRACT

This study aimed to determine elite and poor performance for each offensive and defensive position group on every physical test performed at the NFL Scouting Combine Descriptive statistics were calculated as well as independent samples t-tests for the 15 position groups relative to their performance in the NFL Scouting Combine tests, including: (a) 40-yard Dash, (b) Vertical Jump, (c) Bench Press, (d) Broad Jump, (e) Three-cone Drill, and (f) Shuttle Run. Each player's height and weight were also analyzed. The 15 position groups include Quarterbacks ($n = 206$), Fullbacks ($n = 64$), Running Backs ($n = 314$), Wide Receivers ($n = 501$), Tight Ends ($n = 187$), Offensive Tackles ($n = 279$), Guards ($n = 198$), Centers ($n = 99$), Cornerbacks ($n = 374$), Free Safeties ($n = 132$), Strong Safeties ($n = 129$), Outside Linebackers ($n = 273$), Inside Linebackers ($n = 162$), Defensive Ends ($n = 280$), and Defensive Tackles ($n = 273$). For each of the 15 position groups, there was at least one NFL Scouting Combine test result that had a statistically significant difference at the $p < .05$ or $p < .001$ level when comparing players who averaged playing in at least 10 games per season versus those that played in less than 10 games. The results of this study provide knowledge of which physical test is essential for each offensive and defensive position and can be helpful for those charged with making player personnel decisions in NFL organizations.

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Athlete evaluation; National Football League; scouting combine; strength and conditioning

The National Invitational Camp, or the National Football League (NFL) Scouting Combine, invites top-ranked NFL prospects to perform physical tests to aid NFL teams in their player-selection process (National Football Scouting, 2021). National Football Scouting, Inc. hosted the first camp in 1982, where 163 NFL prospects attended, and in 2023, the NFL Scouting Combine invited 319 prospects to participate (Around the NFL Staff, 2023; Craig Peters, 2018). Despite medical evaluations still being the main priority of the NFL Scouting Combine, physical tests and interviews are also included to provide a more in-depth analysis of each prospect (National Football Scouting, 2021).

NFL teams use the data from the NFL Scouting Combine to inform their personnel decisions. Given NFL rules that (a) limit the number of roster spots, (b) place a cap on player salaries, and (c) limit the number of draft picks each year, organizations have placed a premium on selecting players who have the desired amount of physical ability to compete on the professional level, both immediately and long-term. This is why every NFL team sends members of their scouting department to the NFL Scouting Combine each year, and most head coaches and general managers will attend each year as well (Maaddi, 2024). Cincinnati Bengals General Manager Duke Tobin described the combine as "One hundred percent necessary, in my opinion. You want the right fits for your football team, and the players want the team that drafts them to be the right fit for them because they're dependent on finding a good fit early in their career" (Maaddi, 2023). In 2023, there were also a record number of 1,625 media members that attended the NFL Scouting Combine, which was more than double the amount of media members compared to the 2010 NFL Scouting Combine (Maaddi, 2023). These data support how the NFL Scouting Combine is viewed as an important aspect of the NFL offseason evaluation process.

Previous research found significant correlations between NFL Scouting Combine test results and NFL player performance (LaPlaca & McCullick, 2020). Physical tests performed at the NFL Scouting Combine include the (a) 40-yard Dash, (b) Vertical Jump, (c) Bench Press, (d) Broad Jump, (e) Three-cone Drill, and (f) Shuttle Run. Each of these tests is used to determine an athlete's physical ability under various task constraints possibly important for football. The *40-yard Dash* tests straight-ahead speed and explosion from a static start. The *Vertical Jump* evaluates vertical lower-body power and strength. The *Bench Press* evaluates upper body endurance and strength. The *Broad Jump* is a test of horizontal lower-body power and strength. The *Three-cone Drill* evaluates players' ability to change direction at top speed. The *Shuttle Run* evaluates a player's lateral explosion and quickness in short areas (Pro Football Reference, 2020). Additional evaluations include measurements such as each athlete's height and weight, medical examinations, and interviews.

Despite previous research supporting the correlations between NFL Scouting Combine test results and NFL player performance, NFL teams are missing the exact test result numbers that signify elite performance at the NFL Scouting Combine. For example, previous research has shown significant correlations for quarterbacks between the *40-yard Dash*, *Broad Jump*, and *Shuttle Run* and games played during their NFL career (LaPlaca & McCullick, 2020). However, these data

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do not reveal the exact test result (e.g., a 4.42 Shuttle Run time), which may indicate that a quarterback will play in many NFL games. Thus, during the evaluation process, scouts, coaches, and general managers do not have the specific information that indicates what times or measurements would best project a player's future value to the team. The personnel from NFL teams may realize that the research supports that the *40-yard Dash* correlates with NFL games played for quarterbacks. However, these personnel do not know the specific 40-yard Dash time that they should be looking for during the evaluation process. Even if an NFL team decides to select a player based on a player's performance relative to other players at their position at that year's NFL Scouting Combine, it may not be enough due to the strength or weakness of that position group for that specific year. This study aimed to determine elite and poor performance for each offensive and defensive position group on every physical test performed at the NFL Scouting Combine.

Methods

To find the NFL Scouting Combine test results data for each offensive and defensive position group, the authors collected and analyzed historical data that would best answer the research question guiding the study. This was a secondary data analysis study where other sources already collected data. The researchers collected NFL Scouting Combine test results from Pro Football Reference (2020). The data from *Pro Football Reference* was an open source online, and this source has been used in previous scholarly research (Kmush et al., 2020; LaPlaca & McCullick, 2020; Mulholland & Jensen, 2016).

NFL Scouting Combine test results data were then collected from 2006–2016. This 10-year timeframe is similar to the timeframe used by Mullholland and Jensen's research that projected NFL tight end and wide receiver performance (Mulholland & Jensen, 2016). To account for at least three years of NFL playing experience since the time the player had been drafted, NFL game participation data were collected between 2006–2018.

Subjects

The subjects in this study included all offensive and defensive players who participated in the NFL Scouting Combine between 2006 and 16 ($N = 3{,}438$). These positions included Quarterback ($n = 206$), Fullback ($n = 31$), Running Back ($n = 314$), Wide Receiver ($n = 501$), Tight End ($n = 187$), Offensive Tackle ($n = 279$), Guard ($n = 198$), Center ($n = 99$), Cornerback ($n = 374$), Free Safety ($n = 132$), Strong Safety ($n = 129$), Outside Linebacker ($n = 273$), Inside Linebacker ($n = 162$), Defensive End ($n = 280$), and Defensive Tackle ($n = 273$). Every player's offensive or defensive position was identified based on the position they were labeled at the NFL Scouting Combine. It was deemed by the author's home institution's IRB that since the research did not involve human subjects and the data were obtained from publicly available sources, review and approval were not required.

Data collection procedures

The researchers collected NFL Scouting Combine test results from Pro Football Reference (Kmush et al., 2020). Each NFL Scouting Combine test is described in Table 1. The data from Pro Football Reference was an open source online, and this source has been used in previous studies related to this topic (Mulholland & Jensen, 2016; Parekh & Patel, 2017). Further support for using Pro Football Focus and Pro Football Reference resulted from the data being available for a prolonged period.

Data analysis

The NFL Scouting Combine test results were organized by offensive and defensive positions. The offensive positions were Quarterback, Fullback, Running Back, Wide Receiver, Tight End, Offensive Tackle, Guard, and Center. The defensive positions were Cornerback, Free Safety, Strong Safety, Outside Linebacker, Inside Linebacker, Defensive End, and Defensive Tackle.

All data for each position were then placed individually into a Microsoft Excel document and grouped based on the average number of games played during each player's career from 2006–2018. Specifically, players were grouped based on if they participated in an average of at least or fewer than 10 games per season between 2006–2018. Given that the average NFL player's career is 3.3 years, selecting this timeframe allowed each NFL draft class player to play in the NFL for at least three years after a team signed them (Nguyen, 2022). Only seasons where the player was on an active roster were included in these calculations. For example, if a player only

Table 1. NFL scouting combine tests definitions.

Term	Definition
NFL Scouting Combine Tests	
40-yard Dash	Run 40 yards as fast as possible from a static start. A test of lower body strength and power.
Bench Press	Bench press repetitions of 225 pounds to exhaustion. A test of upper body strength and endurance.
Broad Jump	Jump as far forward as possible. A test of lower body strength and power, primarily occurring through the hips.
Height	Measurement in feet and inches of the athlete's total height from the ground to the top of the athlete's head.
Shuttle Run	Run five yards in one direction, 10 yards in the opposite direction, and then back through the starting line. A test of an athlete'slateral quickness and lower body power.
Three Cone Drill	Athlete performs an agility drill where three cones are set up five yards apart from each other in a "L-shape." A test of change ofdirection at a high speed.
Vertical Jump	Jump as high as possible. A test of lower body strength and power, primarily occurring through bending at the knees.
Weight	Measurement in pounds and ounces of the athlete's total body weight.

played from 2010–2012 and then retired from the NFL, the average was taken based on the number of games they participated in only from 2010–2012.

The 10-game cutoff for the number of games participated in per season was selected because if a player participated in this number of games, it shows that they played in more than half of the games during a season, which signifies the majority of the regular season. It is also worth noting that between the seasons 2006–2018, 16 NFL regular season games were played per season. Still, due to the high occurrence of games missed due to injury in the NFL, it was decided that analyzing players based on whether or not they played in all 16 games would be too high due to the high probability that they would miss at least one game during the season due to injury.

For each group of players, averages, standard deviations, and ranges were calculated for each NFL Scouting Combine testing category. The variables assessed for each position group were *height* (inches), *weight* (pounds), *40-yard Dash* (seconds), *Vertical Jump* (inches), *Bench Press* (repetitions of 225 pounds), *Broad Jump* (inches), *Three-Cone Drill* (seconds), and *Shuttle Run* (seconds), as these are the standard evaluation metrics collected at the NFL Scouting Combine (Pro Football Reference, 2020). The data were then compared between groups to determine which group displayed superior performance in each NFL Scouting Combine testing category. To do this, the difference was calculated between the average values for each group for every NFL Scouting Combine test. The results from this current study were then placed into Microsoft Excel documents, where the data were sorted by offensive and defensive position groups.

In addition, independent samples t-tests were performed in order to determine if there was a significant difference in the NFL Scouting Combine test results between the players that averaged playing in at least 10 games per season versus those that played in fewer than 10 games per season. IBM SPSS was utilized in order to calculate these results. The variables of *height, weight, 40-yard Dash, Vertical Jump, Bench Press, Broad Jump, Three-Cone Drill*, and *Shuttle Run* were inserted into IBM SPSS and sorted into two groups based on if the player had played in at least 10 games per season, or if they averaged playing in fewer than 10 games per season. An independent sample t-test was run in order to determine if any significant differences existed between the two groups. The results for each of the 15 position groups were exported into Microsoft Word documents.

Results

This study endeavored to provide the data that identified elite and poor performance for each offensive and defensive position group for every physical test performed at the NFL Scouting Combine. The analysis of 10 years of NFL Scouting Combine test results and 13 years of NFL playing performance data yielded the mean, standard deviation, and range of performance values for players that played an average of at least 10 games per season between 2006–2018, as well as the mean, standard deviation, and range of performance values for players that played in fewer than 10 games per season between 2006–2018, which are displayed in Tables 2–4. Independent samples t-tests were also performed, comparing players that played in at least 10 games per season versus those that played in fewer than 10 games per season between 2006–2018, and these data are presented in Tables 5 and 6. For each of the 15 position groups, there was at least one NFL Scouting Combine test result that had a statistically significant difference at the $p < .05$ and $p < .001$ levels when comparing players who averaged playing in at least 10 games per season versus those that played in less than 10 games per season.

Discussion

The results of this study revealed the averages, as well as the ranges, and the statistically significant differences that existed between the NFL Scouting Combine test results for players who played an average of 10 or more games per year during their first three seasons in the NFL and those who played an average of fewer than 10 games during their first three seasons in the NFL. The distinction between these two groups is vital because of each team's need to draft players who can contribute to the team's success. The more games a draft pick plays, the more value that player has for the team. These results can be associated with previous research that has found a significant correlation between specific NFL Scouting Combine test results and NFL player performance for each offensive and defensive position group.

Offensive position groups

Quarterback

For the position of Quarterback, previous research has found significant correlations between a better *40-yard Dash*, *Broad Jump*, and *Shuttle Run* with more games played (LaPlaca & McCullick, 2020). This is similar to the current study, where players who participated in more games per season had a statistically significant greater performance in the *Broad Jump*. Previous research has also found that the *Bench Press* produced high correlation values, yet non-significant, due to the low sample size of Quarterbacks that performed the test (LaPlaca & McCullick, 2020). Hedlund (NFL Combine Results, 2020) found that the *Vertical Jump* correlated with more Pro Bowl and All-Pro selections, whereas, in the current study, top performers in this category played in more games per season on average. It is worth noting that in our current study there was clearly a greater range in times for both the *40-yard Dash* and the *Shuttle Run* for players that averaged playing in fewer than 10 games per season compared to players that played in at least 10 games per season. This can potentially speak to the different styles of play that quarterbacks utilize, where some quarterbacks may be more traditional "pocket-passers" that do not accumulate as many rushing yards, compared to "dual-threat" quarterbacks that may look to accumulate more rushing yards (Varshneya, 2023). When quarterbacks are "dual-threat" quarterbacks this can also leave them more susceptible to injury because they are being tackled more frequently than the traditional "pocket-passers" (Varshneya, 2023).

Table 2. Offensive NFL scouting combine performances related to NFL playing experience.

	Quarterback		Fullback		Running Back		Wide Receiver		Tight End		Offensive Tackle		Guard		Center	
	10+	<10	10+	<10	10+	<10	10+	<10	10+	<10	10+	<10	10+	<10	10+	<10
N	20	186	23	41	113	201	186	315	101	86	105	174	56	143	34	65
Height																
n	20	186	23	41	113	201	186	315	101	86	105	174	55	143	34	65
Mean	75.65	74.91	72.7	73.03	70.79	70.36	72.7	72.99	76.67	76.1	77.70	77.45	75.88	76.01	75.41	75.14
SD	1.63	1.56	1.49	1.31	1.94	1.86	2.14	2.3	1.39	1.27	1.21	1.09	1.03	1.25	0.99	1.14
Min	71	71	70	71	66	66	67	65	74	73	75	75	74	74	73	73
Max	78	79	76	75	75	75	78	78	80	79	82	80	78	80	78	78
Weight																
n	20	186	23	41	113	201	186	315	101	86	105	174	55	143	34	65
Mean	228.10	223.13	242.87	244.29	214.98	212.50	202.65	202.39	253.95	252.13	316.17	313.31	314.41	315.69	304.44	302.74
SD	13.52	11.21	8.01	10.25	13.03	13.18	15.79	14.64	8.21	9.58	11.20	10.68	9.40	13.59	6.85	7.40
Min	204	195	226	222	174	168	156	149	230	226	296	277	301	293	280	286
Max	265	260	257	271	247	245	240	238	270	271	358	344	336	364	315	320
40 YD																
n	20	184	23	41	113	201	185	315	100	86	105	174	55	143	34	65
Mean	4.75	4.83	4.76	4.74	4.51	4.57	4.48	4.52	4.73	4.78	5.19	5.25	5.24	5.32	5.21	5.24
SD	0.14	0.18	0.12	0.12	0.10	0.12	0.1	0.09	0.14	0.13	0.17	0.17	0.17	0.18	0.16	0.15
Min	4.52	4.33	4.58	4.53	4.24	4.31	4.26	4.3	4.38	4.4	4.72	4.71	4.91	4.9	4.83	4.93
Max	4.97	5.37	5.06	5.02	4.77	4.84	4.7	4.85	5.06	5.07	5.59	5.75	5.61	6	5.58	5.55
Vertical																
n	15	159	21	34	96	170	157	265	79	64	86	141	41	110	29	49
Mean	31.73	30.95	32.1	33.94	34.58	34.44	35.50	34.93	33.27	32.56	28.26	27.57	27.76	26.43	27.59	26.98
SD	2.97	3.37	2.28	3.39	2.85	3.09	3.06	3.18	3.28	3.38	3.01	3.26	2.81	3.19	2.58	2.48
Min	27	21.5	27.5	28	26.5	25.5	26	26.5	26.5	25	21	19.5	22.5	17.5	23	21
Max	36	40	36	41	41.5	43	45	43.5	42.5	43.5	35.5	35	34	34	33	32.5
Bench																
n	1	16	20	35	95	169	120	193	82	69	91	139	46	121	28	58
Mean	23	19	25.9	21.77	20.33	19.78	14.96	14.73	21.23	20.59	25.52	24.40	27.26	25.63	26.61	26.88
SD	N/A	4.27	5.18	4.75	4.81	4.05	4.10	4.11	4.27	5.09	4.73	5.00	5.87	5.24	4.49	5.14
Min	23	11	16	13	8	8	6	4	13	3	16	9	9	12	18	13
Max	23	26	36	30	32	31	27	27	34	35	38	36	42	45	36	41
Broad																
n	15	161	21	34	95	169	153	260	77	62	82	135	40	109	29	49
Mean	113.87	110.09	111.71	115.03	119.52	117.3	121.22	119.69	115.52	114.53	103.66	101.47	103.00	98.52	101.97	100.53
SD	6.75	6.51	4.94	6.25	5.55	5.63	5.63	5.24	5.29	5.94	6.62	6.77	6.05	6.98	5.11	5.91
Min	103	91	101	105	104	103	103	107	107	101	89	74	91	82	93	89
Max	126	126	119	134	132	135	139	136	131	128	118	117	114	113	113	113
3-Cone																
n	16	150	18	30	73	124	122	220	70	58	81	123	37	105	29	48
Mean	7.01	7.13	7.23	7.2	6.97	7.05	6.92	6.94	7.10	7.20	7.77	7.84	7.77	7.91	7.61	7.75
SD	0.16	0.24	0.24	0.2	0.18	0.19	0.19	0.2	0.20	0.20	0.29	0.30	0.28	0.31	0.21	0.28
Min	6.8	6.55	6.85	6.73	6.5	6.66	6.5	6.42	6.72	6.8	7.25	7.06	7.3	7.26	7.22	7.3
Max	7.4	7.97	7.74	7.68	7.37	7.71	7.45	7.51	7.65	7.62	8.36	8.72	8.26	9.04	8.03	8.24
Shuttle																
n	17	153	17	31	74	120	128	230	70	58	82	127	39	110	29	49
Mean	4.28	4.35	4.42	4.37	4.27	4.28	4.22	4.24	4.37	4.42	4.73	4.79	4.73	4.83	4.58	4.71
SD	0.14	0.17	0.14	0.17	0.14	0.15	0.14	0.15	0.16	0.13	0.18	0.18	0.20	0.21	0.19	0.15
Min	4.09	3.98	4.15	4.03	3.93	3.95	3.81	3.9	4.01	4.19	4.33	4.34	4.4	4.44	4.14	4.27
Max	4.51	4.76	4.7	4.72	4.62	4.73	4.53	4.77	4.82	4.7	5.16	5.24	5.16	5.56	4.96	5.04

10+ = Player had an average of at least participating in 10 games per season between 2006–18 in the NFL; <10 = Player had an average of participating in less than 10 games per season, on average, between 2006–18.

Fullback

Only one previous study analyzed the correlations between a Fullback's NFL Scouting Combine test results and NFL player performance (LaPlaca & McCullick, 2020). It was found that Fullbacks with better *Shuttle Run* times and those lighter in *weight* performed better when running the football, specifically in avoiding tackles per attempt and the longest run achieved on a rushing attempt (LaPlaca & McCullick, 2020). When looking at the current study, it was found that Fullbacks who participated in more games per season had a statistically significant greater performance in the *Vertical Jump*, *Bench Press*, and *Broad Jump*. It was also found in this study that there was a statistically significant difference in *height* between Fullbacks that participated in at least 10 games per season or less than 10 games per season.

Running back

Previous studies found a correlation between a Running Back's performance in the *40-yard Dash* and *Three-cone Drill* at the NFL Scouting Combine and improved NFL player performance (Kuzmits & Adams, 2008; NFL Combine Results, 2020; Teramoto et al., 2016). In this current study, Running Backs that averaged more games played during their career had a statistically significant greater performance in the *40-yard Dash*, *Broad Jump*, and *Three-cone Drill*. In addition, significant, yet weaker, correlations were found in previous research where running backs shorter in *height* and lighter in *weight* were superior in their NFL rushing performance, specifically in more avoided tackles per attempt (LaPlaca & McCullick,

Table 3. Defensive NFL scouting combine performances related to NFL playing experience.

	Cornerback		Free Safety		Strong Safety		Outside Linebacker		Inside Linebacker		Defensive End	Defensive Tackle		
	10+	<10	10+	<10	10+	<10	10+	<10	10+	<10	10+	<10	10+	<10
N	129	246	45	87	42	87	104	169	53	109	131	149	135	138
Height														
n	129	245	45	87	42	87	104	169	53	109	131	149	135	138
Mean	71.26	71.31	72.69	72.57	72.36	72.36	74.16	73.73	73.57	73.28	76.0	75.6	74.92	74.72
SD	1.69	1.63	1.44	1.47	1.50	1.56	1.28	1.38	1.03	1.30	1.5	1.4	1.30	1.38
Min	67	68	70	68	70	69	71	69	71	69	71	73	73	73
Max	75	75	77	76	75	76	78	78	76	77	80	80	78	79
Weight														
n	129	245	45	87	42	87	104	169	53	109	131	149	135	138
Mean	193.52	192.77	206.98	204.52	211.71	210.41	243.79	239.99	242.70	240.68	270.70	266.20	309.41	305.35
SD	8.49	8.51	7.90	8.82	9.99	7.93	11.19	9.25	7.21	6.29	13.46	12.81	14.69	15.81
Min	169	170	195	183	191	184	218	218	230	224	243	235	277	268
Max	219	218	231	230	239	229	270	267	265	258	310	299	354	369
40 YD														
n	129	245	45	87	42	86	104	168	53	108	130	149	135	137
Mean	4.46	4.51	4.54	4.57	4.54	4.58	4.64	4.71	4.70	4.78	4.81	4.85	5.07	5.12
SD	0.09	0.09	0.10	0.10	0.08	0.10	0.11	0.12	0.12	0.13	0.13	0.14	0.17	0.16
Min	4.28	4.29	4.33	4.31	4.34	4.34	4.41	4.4	4.44	4.42	4.53	4.53	4.68	4.78
Max	4.75	4.73	4.75	4.83	4.69	4.85	4.93	5.07	5.05	5.09	5.23	5.23	5.71	5.54
Vertical														
n	112	204	40	74	36	72	93	135	42	86	107	115	110	112
Mean	35.98	35.57	35.60	35.11	35.96	34.68	34.96	33.69	33.21	33.22	33.13	32.35	29.76	28.72
SD	2.57	2.95	2.61	3.47	2.27	2.68	3.42	3.50	3.58	2.91	3.20	3.33	2.93	3.29
Min	30	28.5	30.5	28	32	28	26.5	25	24.5	25.5	25.5	24.5	20.5	20.5
Max	44.5	45	41	43	40.5	40.5	42	42.5	40.5	39.5	42	42	37	36
Bench														
n	110	190	37	73	36	69	84	140	41	90	106	116	105	115
Mean	15.35	14.86	16.22	16.03	18.47	17.38	22.61	22.99	23.32	21.96	24.60	23.26	29.37	27.21
SD	4.11	4.25	3.01	4.30	4.42	4.81	4.85	4.94	3.82	4.07	5.02	4.97	5.35	5.20
Min	4	2	11	7	9	7	11	12	13	11	14	12	17	15
Max	27	26	22	27	29	28	35	41	35	33	38	45	49	44
Broad														
n	114	201	38	72	37	70	93	143	43	89	105	117	105	106
Mean	122.04	121.45	121.16	119.85	121.41	119.21	119.22	116.10	114.63	114.80	115.32	113.56	105.31	104.57
SD	5.70	5.56	6.22	5.56	5.36	5.38	5.76	5.98	7.26	5.22	5.77	6.16	5.51	6.11
Min	109	108	106	108	109	108	107	100	83	104	102	104	90	88
Max	147	135	134	132	132	134	139	134	127	124	129	138	118	118
3-Cone														
n	91	156	32	58	27	58	80	115	31	63	97	104	97	95
Mean	6.92	6.92	6.96	7.06	6.90	7.01	7.09	7.15	7.14	7.15	7.25	7.3	7.66	7.71
SD	0.20	0.20	0.21	0.19	0.23	0.20	0.19	0.24	0.20	0.22	0.23	0.2	0.28	0.28
Min	6.44	6.5	6.58	6.6	6.47	6.55	6.67	6.64	6.73	6.68	6.78	6.9	7.07	7.1
Max	7.31	7.44	7.39	7.49	7.43	7.55	7.55	7.97	7.69	7.68	7.95	7.91	8.28	8.29
Shuttle														
n	97	159	31	58	30	59	80	117	32	69	97	104	97	97
Mean	4.15	4.19	4.20	4.25	4.20	4.25	4.30	4.33	4.26	4.34	4.4	4.5	4.64	4.67
SD	0.13	0.14	0.14	0.14	0.14	0.13	0.14	0.15	0.12	0.16	0.2	0.2	0.16	0.20
Min	3.82	3.89	3.85	3.81	3.9	3.9	3.96	3.96	4	4	4.05	4	4.12	4.21
Max	4.43	4.76	4.46	4.72	4.54	4.51	4.63	4.72	4.56	4.75	4.88	5	5.03	5.27

10+ = Player had an average of at least participating in 10 games per season between 2006–18 in the NFL; <10 = Player had an average of participating in less than 10 games per season, on average, between 2006–18.

2020). Also, those shorter in height had more yards per attempt (LaPlaca & McCullick, 2020). Previous research has also found that the *Broad Jump* and *40-yard Dash* were significantly correlated with more games played for Running Backs (LaPlaca & McCullick, 2020).

Wide Receiver

Three studies found that *Vertical Jump* performance was significantly correlated with NFL player performance for Wide Receivers (Kuzmits & Adams, 2008; NFL Combine Results, 2020; Teramoto et al., 2016). Previous research found a weaker, yet significant, correlation between a better *Shuttle Run* and better offensive and pass-receiving performance (LaPlaca & McCullick, 2020). Compared to the current study, Wide Receivers who averaged playing in more than 10 games per season had a statistically significant greater performance in the *40-yard Dash* and *Broad Jump*.

Tight End

Mulholland and Jensen (2016) found correlations between a Tight End's *Broad Jump* and their NFL performance, which was defined as calculating receiving yards and receiving touchdowns. Hedlund (2018) found a relationship between multiple variables for Tight Ends and success in the NFL, which included the *40-yard Dash* and *Vertical Jump*. Success in the NFL was defined by Hedlund (2018) as having more Pro Bowl or All-Pro selections. LaPlaca and McCullick (2020) also found

Table 4. Offensive and defensive differences in mean NFL scouting combine results between elite and poor performers.

	QB	FB	RB	WR	TE	OT	G	C	CB	FS	SS	OLB	ILB	DE	DT
Height	0.74	−0.33	0.43	−0.29	0.57	0.25	−0.13	0.27	−0.05	0.11	0.00	0.44	0.28	0.39	0.20
Weight	4.97	−1.42	2.48	0.25	1.82	2.86	−1.28	1.70	0.75	2.46	1.30	3.80	2.02	4.50	4.06
40 YD	−0.08*	0.01	−0.06*	−0.04*	−0.05*	−0.06*	−0.08*	−0.03*	−0.05*	−0.03*	−0.04*	−0.07*	−0.07*	−0.04*	−0.05*
Vertical	0.78*	−1.85	0.14*	0.57*	0.71*	0.69*	1.34*	0.61*	0.41*	0.49*	1.28*	1.27*	0.00	0.78*	1.04*
Bench	4.00*	4.13*	0.54*	0.23*	0.64*	1.12*	1.62*	−0.27	0.48*	0.19*	1.10*	−0.38	1.36*	1.35*	2.16*
Broad	3.77*	−3.32	2.21*	1.52*	0.99*	2.18*	4.48*	1.43*	0.60*	1.31*	2.19*	3.12*	−0.17	1.77*	0.75*
3-Cone	−0.12*	0.04	−0.08*	−0.03*	−0.10*	−0.07*	−0.14*	−0.14*	0.01	−0.09*	−0.12*	−0.06*	−0.01*	−0.08*	−0.05*
Shuttle	−0.06*	0.05	−0.02*	−0.02*	−0.05*	−0.06*	−0.10*	−0.13*	−0.03*	−0.05*	−0.06*	−0.04*	−0.08*	−0.05*	−0.04*

Calculations were made by subtracting the mean value of the respective NFL Scouting Combine result of players that had an average of at least 10 NFL games played per season, from the mean value for the respective NFL Scouting Combine result of players from the same position group that did not average playing 10 NFL games per season. *Indicates that the NFL Scouting Combine performance metric favors the players that played an average of at least 10 NFL games per season. QB=Quarterback; FB=Fullback; RB=Running Back; WR=Wide Receiver; TE=Tight End; OT=Offensive Tackle; G=Guard; C=Center; CB=Cornerback; FS=Free Safety; SS=Strong Safety; OLB=Outside Linebacker; ILB=Inside Linebacker; DE=Defensive End; DT=Defensive Tackle.

that the *Vertical Jump* correlated with a better offensive receiving performance, and that *height* was significantly correlated with more games played. In our current study, it was found that players that averaged playing in more than 10 games per season had a statistically significant greater performance in the *40-yard Dash* and *Three-cone Drill*, as well as had a significant difference in *height*.

Offensive Tackle

LaPlaca and McCullick (2020) was the only previous study to examine the Offensive Tackle position specifically relating to NFL Scouting Combine test results and NFL player performance. However, Hedlund (2018) studied the position of "offensive line" in general, which grouped Offensive Tackles, Guards, and Centers. LaPlaca and McCullick (2020) found a robust and significant correlation between having a better *Bench Press* and *40-yard Dash* and a better NFL performance, explicitly relating to pass blocking. LaPlaca and McCullick (2020) found that Offensive Tackles shorter in *height* had better NFL performance, specifically related to having fewer sacks and hurries allowed. Previous research has also found that the *40-yard Dash* was significantly correlated with more games played for Offensive Tackles (LaPlaca & McCullick, 2020). LaPlaca and McCullick (2020) found that even though Offensive Tackles, Guards, and Centers all play on the offensive line, the study results for each position group varied greatly. This current study also found that elite-level NFL performers at the Offensive Tackle position had statistically significant better performance in the *40-yard Dash* and *Three-cone Drill*. There was also a statistically significant difference in *weight*.

Guard

To this point, there has only been one previous study that measured the correlations between NFL Scouting Combine test results and NFL player performance for Guards, and only weak significant correlations were found (LaPlaca & McCullick, 2020). Previous research has found that the *Vertical Jump* and *Broad Jump* were significantly correlated with more games played by Guards (LaPlaca & McCullick, 2020). The *Shuttle Run* was found to be correlated with better overall offensive NFL performance. In our current study found that those players that played in more games had a statistically significant difference in performance in the *40-yard Dash*, *Vertical Jump*, *Broad Jump*, *Three-cone Drill*, and the *Shuttle Run*.

Center

As with Offensive Tackles and Guards, only one previous study researched Centers specifically (LaPlaca & McCullick, 2020). Like Guards, the *Shuttle Run* indicated better NFL player performance regarding pass blocking (LaPlaca & McCullick, 2020). Previous research has found that the *Shuttle Run* was significantly correlated with more games played for Centers (LaPlaca & McCullick, 2020). The current study's results agree with previous research that guards with more NFL experience had statistically significant different results in the *Three-cone Drill* and the *Shuttle Run* compared to other players who did not play in as many NFL games.

Defensive position groups

Cornerback

While Hedlund (2018) studied Defensive Backs in general, LaPlaca and McCullick (2020) was the only previous study to look at the position of Cornerback specifically. It was found that having a better *40-yard Dash* was correlated with more games played during their career and more pass breakups. Previous research has found that the *40-yard Dash* and *Shuttle Run* were significantly correlated with more games played for cornerbacks (LaPlaca & McCullick, 2020). Similar to previous research, the current study also found that elite-level Cornerbacks had a statistically significant difference in performance in the *40-yard Dash* compared to those players who did not participate in as many games during their NFL careers.

Free safety

Similar to the Cornerback position, only one previous study looked specifically at the position of Free Safety (LaPlaca & McCullick, 2020). LaPlaca and McCullick (2020) found that superior *Vertical Jump* and *Broad Jump* strongly correlated with more tackles per snap count. Weaker yet significant correlations found that the *Bench Press* and *Three-cone Drill* were associated with having better NFL performance (LaPlaca & McCullick, 2020). The *40-yard Dash* was associated with more interceptions and pass breakups per snap count (LaPlaca & McCullick, 2020). Results from previous research align with our current study, where elite-level Free Safeties were found to have a statistically significant difference in performance in the *Three-cone Drill* compared to players that did not play in as many games.

Table 5. Offensive Independent samples *t*-test.

	Quarterback	Fullback	Running Back	Wide Receiver	Tight End	Offensive Tackle	Guard	Center
Height								
t	2.016	−.779	1.935	−1.422	2.909	1.782	−.748	1.181
df	204	62	312	499	185	277	196	97
Two-Sided p	.045*	.439	.054	.156	.004*	.076	.456	.241
Mean Difference	.741	−.280	.429	−.295	.569	.251	.134	.273
SE Difference	.368	.359	.222	.207	.195	.141	.180	.231
Weight								
t	1.843	−.574	1.607	.183	1.401	2.128	−.498	1.115
df	204	62	312	499	185	277	196	97
Two-Sided p	.067	.568	.109	.855	.163	.034*	.619	.268
Mean Difference	4.966	−1.423	2.480	.255	1.823	2.861	−.983	1.703
SE Difference	2.694	2.480	1.544	1.394	1.301	1.345	1.973	1.528
40 YD								
t	−1.915	.442	−4.284	−4.526	−2.630	−2.709	−3.000	−1.088
df	202	62	312	498	184	277	196	97
Two-Sided p	.057	.660	<.001**	<.001**	.009*	.007*	.003*	.279
Mean Difference	−.080	.014	−.056	−.040	−.051	−.058	−.083	−.035
SE Difference	.042	.031	.013	.009	.019	.022	.028	.032
Vertical								
t	.869	−2.204	.372	1.814	1.269	.878	2.273	1.028
df	172	53	264	420	141	225	149	76
Two-Sided p	.386	.032*	.710	.070	.207	.381	.024*	.307
Mean Difference	.784	−1.846	.143	.573	.710	.422	1.282	.607
SE Difference	.901	.838	.384	.316	.559	.481	.564	.590
Bench								
t	.908	2.999	.932	.488	.837	1.698	1.946	−.239
df	15	53	241	311	149	228	165	84
Two-Sided p	.378	.004*	.352	.626	.404	.091	.053	.811
Mean Difference	4.000	4.129	.542	.233	.638	.1.121	1.796	−.272
SE Difference	4.405	1.376	.582	.478	.762	.660	.923	1.137
Broad								
t	2.141	−2.063	3.081	2.776	1.035	2.324	3.522	1.089
df	174	53	262	411	137	215	147	76
Two-Sided p	.034*	.044*	.002*	.006*	.303	.021*	<.001**	.280
Mean Difference	3.773	−3.315	2.214	1.523	.987	2.184	4.269	1.435
SE Difference	1.763	1.607	.719	.549	.954	.940	1.212	1.318
3-Cone								
t	−1.896	.577	−2.942	−1.177	−2.829	−1.644	−2.161	−2.261
df	164	46	195	340	126	202	140	75
Two-Sided p	.060	.567	.004*	.240	.005*	.102	.032*	.027*
Mean Difference	−.118	.037	−.080	−.026	−.103	−.070	−.129	−.136
SE Difference	.062	.064	.027	.022	.036	.042	.060	.060
Shuttle								
t	−1.470	.968	−.861	−1.216	−1.750	−2.263	−2.299	−3.344
df	168	46	192	356	126	207	147	76
Two-Sided p	.143	.338	.390	.225	.083	.025*	.023*	.001*
Mean Difference	−.061	.046	−.019	−.019	−.046	−.059	−.088	−.131
SE Difference	.042	.048	.022	.016	.026	.026	.038	.039

t = t-score; df = degrees of freedom; SE = Standard Error. * = Sig. at $p < .05$; ** = Sig. at $p < .001$.

Strong safety

The correlation between a Strong Safety's NFL Scouting Combine test results and their NFL player performance has only been researched in one previous study (LaPlaca & McCullick, 2020). All NFL player performance indicators correlated with *height* and *weight* favored taller and heavier strong safeties, primarily for their ability to tackle. Interestingly, although weakly correlated, a better *Bench Press* was associated with a better defense against the run. In contrast, a better *Three-cone Drill* and *Shuttle Run* were the two NFL Scouting Combine test results related to better pass coverage. Previous research has found that the *Vertical Jump* was significantly correlated with more games played for Strong Safeties (LaPlaca & McCullick, 2020). In our current study, it was found that Strong Safeties that averaged playing in more games during their NFL career had a statistically significant difference in performance in the *40-yard Dash*, *Vertical Jump*, *Broad Jump*, and *Three-cone Drill*.

Outside Linebacker

LaPlaca and McCullick's (2020) study was the first to look at the Outside Linebacker position specifically, however previous studies have looked at the Linebacker position in general (Hedlund, 2018; Teramoto et al., 2016). Vincent, Blissmer, and Hatfield (Teramoto et al., 2016) found that Linebackers with a better *40-yard Dash* had more tackles and sacks, those with a higher *Vertical Jump* had more sacks and defended passes, and those with a better *Shuttle Run* had more sacks. These results are similar to LaPlaca and McCullick's (2020), where the *40-yard Dash* and *Shuttle Run* were significantly correlated to NFL player performance indicators for Outside Linebackers. LaPlaca and McCullick

Table 6. Defensive Independent samples *t*-test.

	Cornerback	Free Safety	Strong Safety	Outside Linebacker	Inside Linebacker	Defensive End	Defensive Tackle
Height							
t	−.282	.426	.003	2.600	1.381	2.217	1.238
df	372	130	127	271	160	278	271
Two-Sided p	.778	.671	.998	.010*	.169	.027	.217
Mean Difference	−.051	.114	.001	.436	.282	.386	.201
SE Difference	.180	.268	.289	.168	.204	.174	.162
t	.809	1.573	.800	3.040	1.827	2.865	2.197
Weight							
df	372	130	127	271	160	278	271
Two-Sided p	.419	.118	.425	.003*	.070	.004*	.029*
Mean Difference	.748	2.461	1.300	3.800	2.019	4.501	4.060
SE Difference	.925	1.564	1.625	1.25	1.105	1.571	1.848
40 YD							
t	−5.119	−1.672	−2.239	−4.582	−3.412	−2.541	−2.410
df	372	130	126	270	159	277	270
Two-Sided p	<.001**	.097	.027*	<.001**	<.001**	.012*	.017*
Mean Difference	−.050	−.031	−.040	−.068	−.072	.-.040	−.048
SE Difference	.010	.019	.018	.015	.021	.016	.020
Vertical							
t	1.241	.773	2.454	2.719	−.001	1.772	2.478
df	314	112	106	226	126	220	220
Two-Sided p	.216	.441	.016*	.007*	.999	.078	.014*
Mean Difference	.412	.485	1.278	1.270	−.001	.778	1.036
SE Difference	.332	.628	.521	.467	.592	.439	.418
Bench							
t	.959	.239	1.137	−.559	1.809	2.004	3.040
df	298	108	103	222	129	220	218
Two-Sided p	.339	.812	.258	.577	.073	.046*	.003*
Mean Difference	.482	.189	1.095	−.379	1.362	1.345	2.163
SE Difference	.503	.791	.963	.677	.753	.671	.712
Broad							
t	.906	1.128	2.007	3.969	−.154	2.2	.934
df	313	108	105	234	130	220	209
Two-Sided p	.366	.262	.047*	<.001**	.878	.029*	.351
Mean Difference	.596	1.311	2.191	3.117	−.170	1.768	.748
SE Difference	.658	1.162	1.092	.785	1.106	.804	.801
3-Cone							
t	.224	−2.170	−2.423	−2.014	−.301	−2.484	−1.208
df	245	88	83	193	92	199	190
Two-Sided p	.823	.033*	.018*	.045*	.764	.014*	.229
Mean Difference	.006	−.093	−.118	−.065	−.014	−.082	−.049
SE Difference	.026	.043	.049	.032	.047	.033	.040
Shuttle							
t	−1.887	−1.462	−1.833	−1.688	−2.597	−2.106	−1.380
df	254	87	87	195	99	199	192
Two-Sided p	.060	.147	.070	.093	.011*	.036*	.169
Mean Difference	−.034	−.045	−.056	−.036	−.082	−.053	−.036
SE Difference	.018	.031	.030	.021	.032	.025	.026

t = t-score; df = degrees of freedom; SE = Standard Error. * = Sig. at $p < .05$; ** = Sig. at $p < .001$.

(2020) also found that a better *Three-cone Drill* correlated with better overall defensive performance, specifically in pass coverage. Previous research has found that the *40-yard Dash* and *Broad Jump* were significantly correlated with more games played for Outside Linebackers (LaPlaca & McCullick, 2020). The results from our current study concur with previous research where elite-level Outside Linebackers had statistically significant different results in the *40-yard Dash*, *Vertical Jump*, *Broad Jump*, and *Three-cone Drill* compared to those that did not participate in as many games. There was also a statistically significant difference seen in *height* and *weight*.

Inside Linebacker

As with Outside Linebackers, only one previous study examined the Inside Linebacker position (LaPlaca & McCullick, 2020). The *Bench Press* was the NFL Scouting Combine test result with the greatest number of relevant significant correlations to NFL player performance. The NFL Scouting Combine test results of an Inside Linebacker correlated primarily toward run defense and pass rush. Although weakly correlated, the *Vertical Jump* and *Broad Jump* significantly correlated to the overall defensive performance. Previous research has found that the *40-yard Dash, Bench Press*, and *Shuttle Run* were significantly correlated with more games played for Inside Linebackers (LaPlaca & McCullick, 2020). Similar to previous research, it was found in our current study that Inside Linebackers that averaged more games played had a statistically significant difference in performance in the *40-yard Dash* and the *Shuttle Run*.

Defensive End

Previous studies found significant correlations between a Defensive End's NFL Scouting Combine test results and

their NFL player performance (Hedlund, 2018; Vincent et al., 2019). Vincent, Blissmer, and Hatfield's (Vincent et al., 2019) study found correlations between *Vertical Jump* scores and total tackles and sacks and also found that the *Shuttle Run* is correlated with the number of sacks for Defensive Ends. LaPlaca and McCullick (2020) found correlations relating a Defensive End's *Vertical Jump* and *Shuttle Run* related to NFL player performance. The *Three-cone Drill*, *Broad Jump*, and *40-yard Dash* test results strongly correlate to a Defensive End's ability to rush the passer. The current study's most highly correlated NFL Scouting Combine test was the *40-yard Dash*, with seven strong correlations relating primarily to the Defensive End's pass-rushing ability. Previous research has found that *weight* was significantly correlated with more games played for Defensive Ends (LaPlaca & McCullick, 2020). Historical research aligns with the results from our current study, where it was found that elite-level Defensive Ends had statistically significant different results in the *40-yard Dash*, *Bench Press*, *Broad Jump*, *Three-cone Drill*, and the *Shuttle Run*. Similar to previous research, it was also found in our current study that NFL players with more games played had a statistically significant different in *weight* compared to players that did not play as much.

Defensive Tackle

Previous studies also found significant correlations between a Defensive Tackle's NFL Scouting Combine test results and their NFL player performance (NFL Combine Results, 2020; Vincent et al., 2019). Studies by Vincent, Blissmer, and Hatfield (Vincent et al., 2019) and LaPlaca and McCullick (2020) specifically found that a Defensive Tackle's *Broad Jump* was significantly correlated with the number of sacks they acquired. LaPlaca and McCullick (2020) also found that a better *40-yard Dash* is associated with a Defensive Tackle's ability to rush the passer, specifically regarding pressures, sacks, hits, and hurries. Previous research has found that *Vertical Jump, Bench Press, Broad Jump, Three-cone Drill*, and *Shuttle Run* were significantly correlated with more games played for Defensive Tackles (LaPlaca & McCullick, 2020). These previous findings agree with our current study where it was found that elite-level Defensive Tackles had statistically significant different results in the *Broad Jump, Vertical* Jump, *Bench Press*, *40-yard Dash* compared to those Defensive Tackles that did not participate in as many games on average. Our current research also found that elite-level performers had a statistically significant difference in *weight*.

Practical applications

The data collected from this study can be used to identify the top talent on display at the NFL Scouting Combine. We now have information to support which skills tests are essential for each position group and what measurements are deemed as elite for each position group for each skill test. Coaches, scouts, and general managers can now use this information during their evaluation process of each prospect to aid in selecting the prospect that has the best chance of succeeding in the NFL. Before this study was conducted, there were no data to support which measurements player personnel should be looking for in each skills test to identify an elite performance. The data from this study provide NFL teams with which measurements may be best used to determine top talent.

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Author's contributions

Dr. David LaPlaca conducted the primary data collection and rough draft of the manuscript. Dr. Bryan McCullick provided guidance during the data collection process and helped with writing the manuscript.

Data availability statement

The data that support the findings of this study are openly available through *Pro Football Reference* at https://www.pro-football-reference.com/, reference number 6.

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